

TASK 2 REPORT

Development and Testing of Custom Correlation Rules in ELK Stack SIEM for Detection of Credential Stuffing, DNS Tunnelling, and PowerShell Exploitation

Report	Task 2 - Development and Testing of Custom Correlation Rules in ELK Stack SIEM
SIEM Platform	ELK Stack
Detection Language	KQL (Kibana Query Language)
Attack Scenarios	Credential Stuffing, DNS Tunnelling, and PowerShell Exploitation

Abstract

Security Information and Event Management (SIEM) systems play a critical role in identifying malicious activities by collecting, analyzing, and correlating security events from multiple sources. This project focuses on the development and testing of custom correlation rules in the ELK Stack (Elasticsearch, Logstash, and Kibana) to detect three common attack techniques: Credential Stuffing, DNS Tunnelling, and PowerShell Exploitation. Simulated attack logs were generated and ingested into Elasticsearch. Custom detection rules were configured in Kibana Security, and alerts were successfully generated upon detecting suspicious activities. The results demonstrate the effectiveness of SIEM-based detection mechanisms in identifying security threats in real time.

1. Introduction

Cybersecurity threats continue to evolve and pose significant risks to organizations. Attackers frequently employ techniques such as credential stuffing, DNS tunnelling, and malicious PowerShell execution to gain unauthorized access and evade traditional security controls.

Security Information and Event Management (SIEM) solutions help security teams monitor, analyze, and detect suspicious activities by aggregating logs from various sources and applying correlation rules.

In this project, the ELK Stack was used as a SIEM platform to develop and test custom correlation rules capable of detecting:

- Credential Stuffing Attacks
- DNS Tunnelling Activities
- PowerShell Exploitation Attempts

2. Objectives

The primary objectives of this project are:

- To deploy and configure the ELK Stack as a SIEM solution.
- To generate simulated attack logs.
- To create custom detection rules using Kibana Query Language (KQL).
- To detect credential stuffing attacks.
- To identify DNS tunnelling activities.
- To monitor suspicious PowerShell execution.
- To generate and validate security alerts.
- To analyze detection results and evaluate rule effectiveness.

3. Tools and Technologies Used

Component	Purpose
Elasticsearch	Log storage and indexing
Kibana	Visualization and security monitoring
ELK Stack	SIEM Platform
KQL (Kibana Query Language)	Rule creation and filtering
Windows PowerShell	Attack simulation
JSON Logs	Event generation and ingestion

4. ELK Stack Architecture

The ELK Stack consists of three major components:

Elasticsearch

Stores and indexes security logs for efficient searching and analysis.

Kibana

Provides dashboards, visualizations, and detection rule management.

Attack Logs Dataset

Custom logs containing simulated attack events were indexed into Elasticsearch under the index:

```
attack-logs
```

The following fields were included:

- @timestamp
- event_type
- source_ip
- username
- status
- dns_query
- powershell_command

5. Attack Simulation

5.1 Credential Stuffing Simulation

Credential stuffing attacks involve repeated login attempts using stolen usernames and passwords.

```
{ "@timestamp": "2026-06-18T05:40:00Z", "event_type": "login", "source_ip": "192.168.1.150",  
  "username": "admin", "status": "failed" }
```

Multiple failed login attempts were generated from different IP addresses and indexed into Elasticsearch.

5.2 DNS Tunnelling Simulation

DNS tunnelling uses DNS requests to covertly transfer data between systems.

```
{ "@timestamp": "2026-06-24T04:25:00Z", "event_type": "dns", "source_ip": "192.168.1.100",  
  "dns_query": "verylongsubdomain.maliciousdomain.com" }
```

DNS queries were inserted into Elasticsearch to simulate data exfiltration behavior.

5.3 PowerShell Exploitation Simulation

PowerShell is frequently abused by attackers to execute malicious commands.

```
{ "@timestamp": "2026-06-25T12:10:00Z", "event_type": "powershell", "source_ip": "192.168.1.250",  
  "powershell_command": "powershell.exe -EncodedCommand ALERTTEST" }
```

Suspicious PowerShell commands were generated and indexed into Elasticsearch.

6. Custom Correlation Rules

6.1 Credential Stuffing Detection Rule

Property	Value
Rule Name	Credential Stuffing Detection
Index Pattern	attack-logs
KQL Query	event_type:"login" AND status:"failed"
Severity	Medium
Risk Score	50

This rule identifies failed login attempts that may indicate credential stuffing attacks. When multiple failed authentication attempts are observed, an alert is generated for further investigation.

6.2 DNS Tunnelling Detection Rule

Property	Value
Rule Name	DNS Tunnelling Detection
Index Pattern	attack-logs
KQL Query	dns_query:*
Severity	High
Risk Score	75

This rule monitors DNS query activity and detects DNS requests that may indicate covert communication channels or data exfiltration through DNS tunnelling.

6.3 PowerShell Exploitation Detection Rule

Property	Value
Rule Name	PowerShell Exploitation Detection
Index Pattern	attack-logs
KQL Query	event_type:"powershell"
Severity	High
Risk Score	75

This rule detects suspicious PowerShell execution events, particularly encoded PowerShell commands often associated with malicious activity.

7. Rule Configuration

All rules were configured with:

- Rule Type: Query Rule
- Schedule Interval: 1 Minute
- Additional Look-back Time: 1 Hour
- Maximum Alerts Per Run: 100

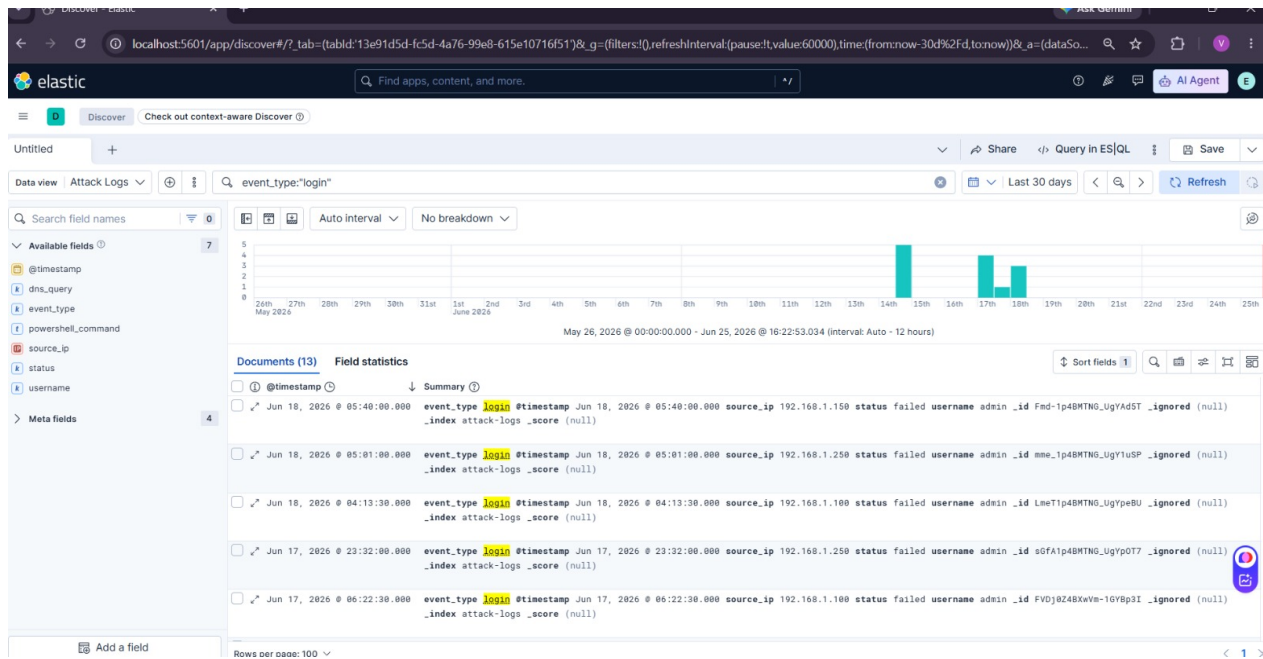
The rules continuously monitor incoming events and generate alerts whenever matching events are detected.

8. Results and Validation

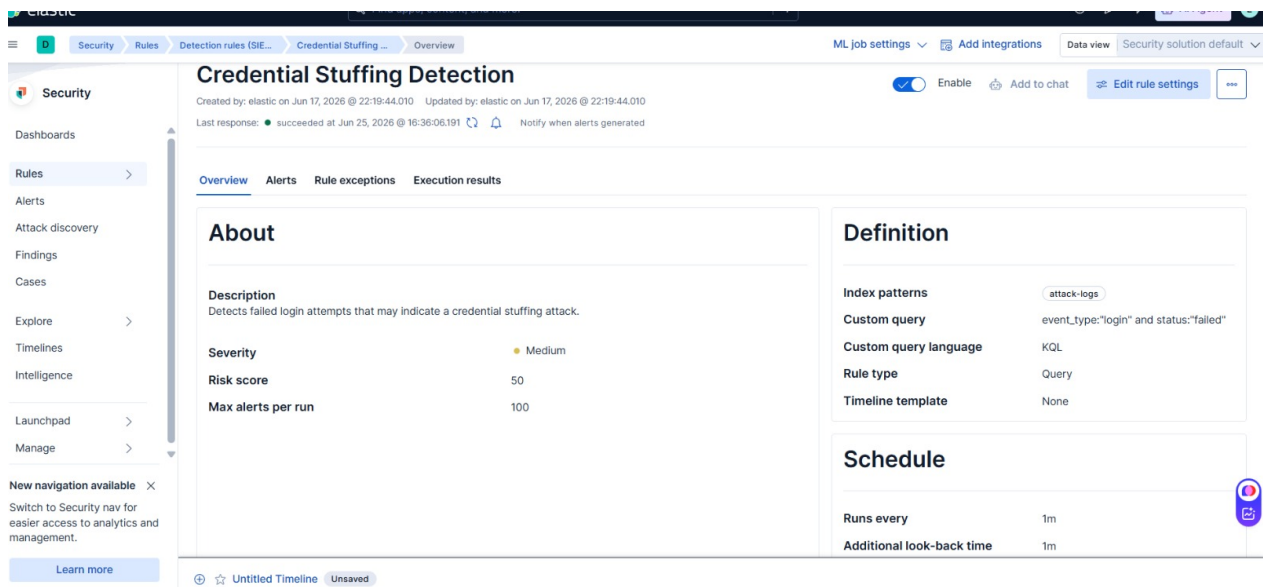
The custom rules successfully detected the simulated attack events.

Credential Stuffing Results

The SIEM generated alerts based on failed login attempts recorded in Elasticsearch.



Screenshot 1: Credential Stuffing Logs from Discover



Screenshot 2: Credential Stuffing Detection Rule

DNS Tunnelling Results

The SIEM successfully identified suspicious DNS query activity and generated alerts.

elastic Find apps, content, and more.

Security Rules Detection rules (SIE...) DNS Tunnelling Dete... Overview ML job settings Add integrations Data view Security solution default

Security

Dashboards

Rules Alerts Attack discovery Findings Cases Explore Timelines Intelligence Launchpad Manage

New navigation available Switch to Security nav for easier access to analytics and management. Learn more

DNS Tunnelling Detection

Created by: elastic on Jun 17, 2026 @ 22:26:56.430 Updated by: elastic on Jun 23, 2026 @ 21:57:16.699

Last response: succeeded at Jun 25, 2026 @ 16:36:00.295 Notify when alerts generated

Overview Alerts Rule exceptions Execution results

About

Description
Detects DNS query events that may indicate DNS tunnelling activity.

Severity Low

Risk score 60

Max alerts per run 100

Definition

Index patterns attack-logs

Custom query dns_query:*

Custom query language KQL

Rule type Query

Timeline template None

Schedule

Runs every 1m

Additional look-back time 1h

Untitled Timeline Unsaved

Screenshot 3: DNS Tunnelling Rule Overview

elastic Find apps, content, and more.

Security Alerts ML job settings Add integrations Data view Security solution default

Filter your data using KQL syntax Last 7 days Refresh

Alerts

Assignees Manage rules

Status Severity User Host

Summary Trend Counts Treemap

Severity levels

Levels	Count
High	1
Medium	1
Low	4

6 alerts

Alerts by name

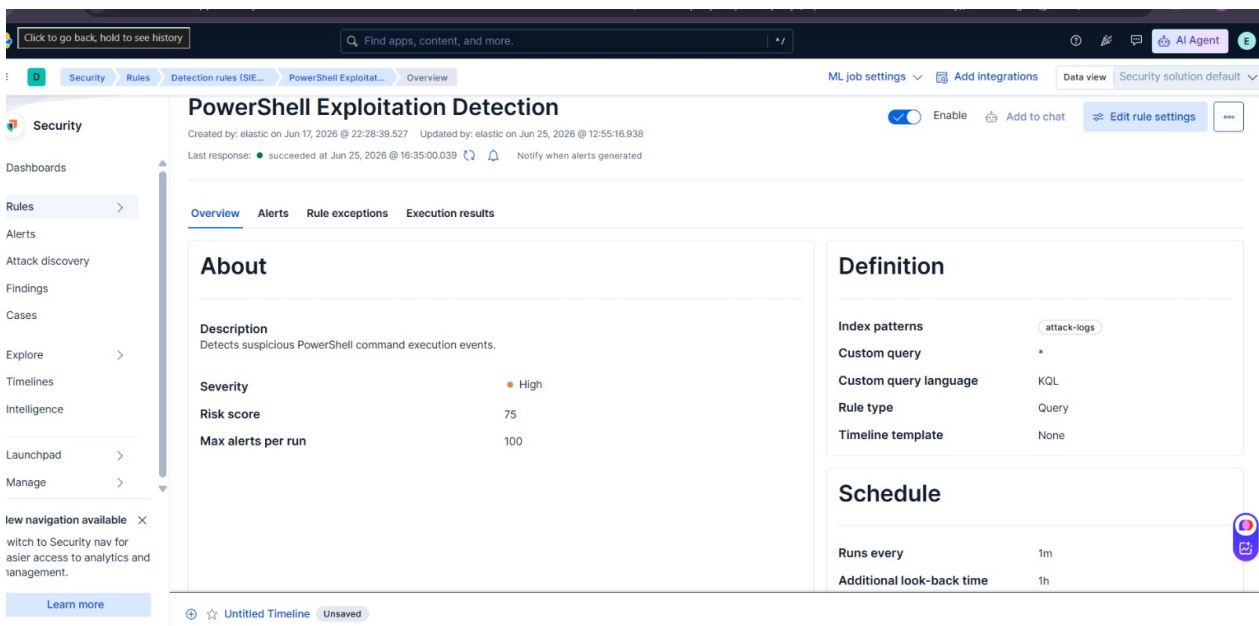
Rule name	Count
DNS Tunnelling Detection	2
Credential Stuffing Detection	1
Failed Login Detection	1
Failed Login Detection [Duplicate]	1

Untitled Timeline Unsaved

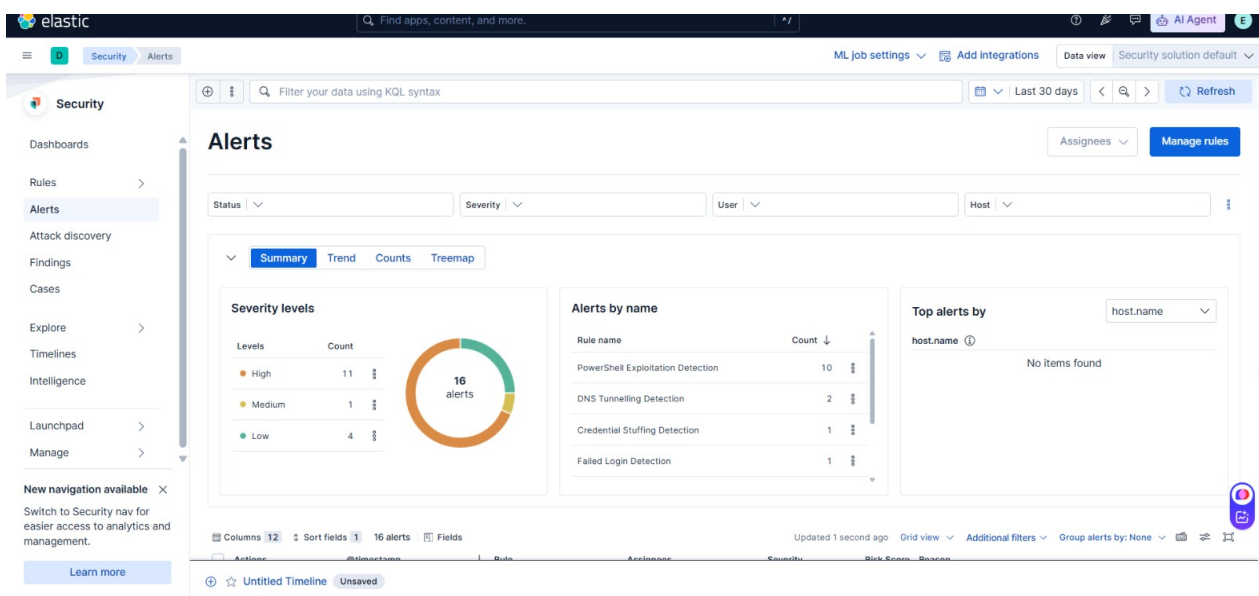
Screenshot 4: DNS Alert Details

PowerShell Exploitation Results

The PowerShell detection rule successfully identified encoded PowerShell commands and generated high-severity alerts.



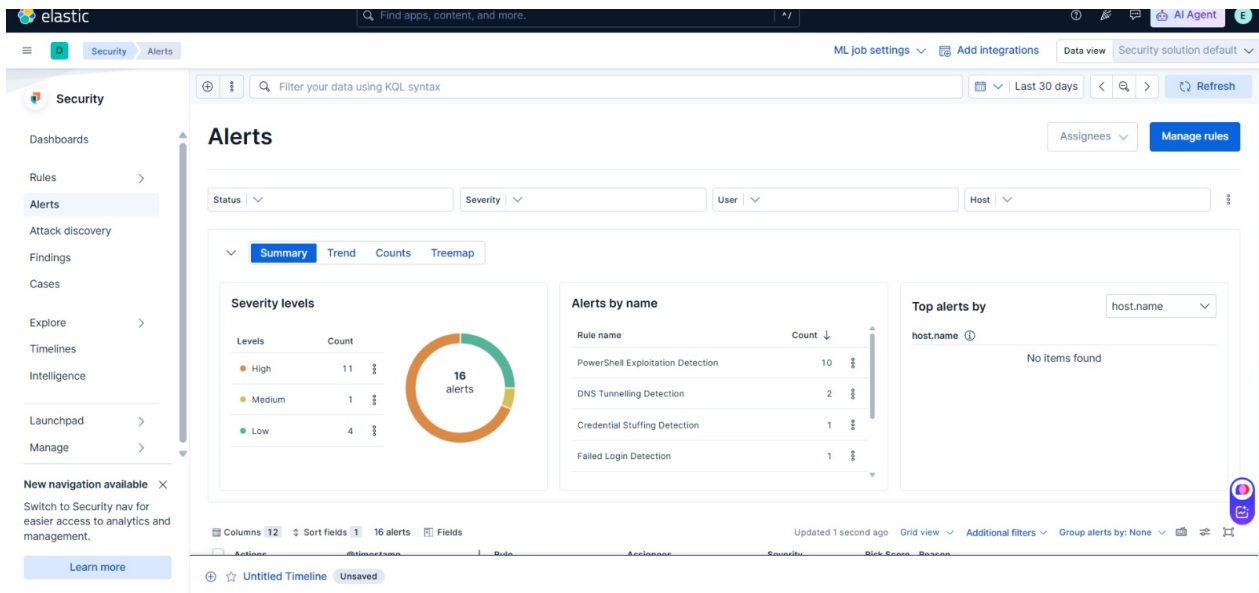
Screenshot 5: PowerShell Rule Overview



Screenshot 6: PowerShell Alert Details

Alert Dashboard Results

The Security Alerts dashboard displayed alerts generated by all custom correlation rules.



Screenshot 7: Security Alerts Dashboard

Observations:

- Total Alerts Generated: 16
- PowerShell Alerts: 10
- DNS Tunneling Alerts: 2
- Credential Stuffing Alerts: 1
- Additional Test Alerts: 3

9. Findings

The implementation demonstrated that:

- ELK Stack can effectively function as a SIEM platform.
- Custom KQL-based correlation rules successfully detected simulated attack patterns.
- Alert generation occurred in near real-time.
- Kibana dashboards provided clear visualization and monitoring capabilities.
- Rule severity and risk scoring helped prioritize security incidents.

10. Challenges Faced

During implementation, several challenges were encountered:

- Time synchronization issues between event timestamps and rule execution windows.
- Data visibility delays due to time filter settings.
- Initial alert generation failures caused by mismatched timestamps.
- Verification of rule execution and event ingestion.

These challenges were resolved by adjusting timestamps and configuring appropriate time ranges in Kibana.

11. Conclusion

This project successfully developed and tested custom SIEM correlation rules using the ELK Stack for detecting Credential Stuffing, DNS Tunnelling, and PowerShell Exploitation attacks. Simulated attack events were ingested into Elasticsearch, and Kibana Security was used to create detection rules and generate alerts. The successful detection and alerting of suspicious activities demonstrate the effectiveness of SIEM solutions in improving organizational security monitoring and incident detection capabilities.

The project highlights how ELK Stack can be leveraged as a cost-effective and scalable SIEM platform for threat detection and security operations.

12. References

- Elastic Documentation – Elasticsearch and Kibana Security.
- MITRE ATT&CK Framework.
- OWASP Credential Stuffing Prevention Guide.
- Microsoft PowerShell Security Documentation.
- DNS Tunnelling Detection Research Papers.